

Post-Exploitation Enumeration and Tools

CompTIA PenTest+ (PT0-003)

Domain: 5.9: Post-exploitation and Lateral Movement

Objectives:

- 5.2: Given a scenario, perform tasks to move laterally throughout the environment.
- 5.1: Given a scenario, perform tasks to establish and maintain persistence.

Video Overview

This video provides an in-depth look at post-exploitation techniques, a vital skill set for anyone pursuing a career in penetration testing or cybersecurity. As you progress in your learning, you'll find that gaining initial access to a system is only the first step. The real challenge, and often the most insightful part of an engagement, comes from understanding how to expand your reach within a compromised network, identify sensitive information, and move laterally to other systems. The concepts and tools discussed here, such as PowerShell Empire, Mimikatz, and BloodHound, are not just theoretical; they are actively used in the field to assess the true depth of a breach and demonstrate its potential impact. By mastering these techniques, you will be better equipped to protect organizations from sophisticated threats, making a tangible difference in real-world security scenarios.

Key Topics Covered

- Post-exploitation techniques for expanding control and identifying sensitive information.
- Enumeration targets: sensitive data (passwords, PII, PHI, R&D schematics) and network mapping (users, groups, domains in Active Directory).
- "Living off the land" approach using built-in system utilities.
- PowerShell Empire: C2 framework for interactive control, including commands like `help`, `agents`, `listeners`, and `interact`.
- PowerShell Empire workflow: setting up listeners, creating stagers, and establishing agent connections.

- Mimikatz: tool for extracting usernames, password hashes, and cleartext passwords from memory.
- BloodHound: tool for Active Directory mapping, visualizing relationships and privilege escalation paths.
- Lateral Movement and Pass-the-Hash technique using stolen password hashes and tools like `pth-winexe`.

Student Notes Section